# AI Performance Benchmark Report – Zecpath

## Objective

Measure and optimize the performance of AI services for production-scale recruitment.

# Test Environment

| Parameter | Value |
|---|---|
| Concurrent Users | 500 |
| Requests / Second | 200 |
| Test Duration | 30 Minutes |

# Performance Metrics

| Metric | Before | After |
|---|---|---|
| Average Response Time | 2.1 s | 0.9 s |
| Maximum Latency | 4.5 s | 1.8 s |
| Throughput | 120 req/s | 260 req/s |
| CPU Usage | 80% | 55% |
| Memory Usage | High | Optimized |

# Optimization Techniques

- LRU caching
- Batch processing
- Generator-based memory optimization
- Faster decision engine
- Optimized API responses

# Results

- 57% reduction in latency
- 2× throughput improvement
- Lower memory consumption

- Improved scalability

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# Conclusion

The optimized AI services are capable of handling significantly higher workloads with lower latency and improved resource utilization.